

```
1 #include "FeetInches.h"
2 #include <string>
3
4 // Constructors
5 FeetInches::FeetInches()
6 {
7     inches = 0;
8     feet = 0;
9 }
10
11 FeetInches::FeetInches(double f, double i)
12 {
13     while (i >= 12)
14     {
15         i -= 12;
16         f++;
17     }
18     inches = i;
19     feet = f;
20 }
21
22 FeetInches::~FeetInches() {}
23
24 // Setters
25 void FeetInches::setInches(double i)
26 {
27     while (i >= 12) {
28         i -= 12;
29         feet++;
30     }
31     inches = i;
32 }
33
34 void FeetInches::setFeet(double f)
35 {
36     feet = f;
37 }
38
39 void FeetInches::setTotal(double f, double i)
40 {
41     while (i >= 12)
42     {
43         i -= 12;
44         f++;
45     }
46     inches = i;
47     feet = f;
48 }
49
```

```
50 // Getters
51 double FeetInches::getInches()const
52 {
53     return inches;
54 }
55
56 double FeetInches::getFeet()const
57 {
58     return feet;
59 }
60
61 // Overloaded Operators
62 void FeetInches::operator +(FeetInches& dis2)
63 {
64     double tempFt = dis2.feet;
65     double tempIn = dis2.inches;
66     while (tempIn >= 12)
67     {
68         tempIn -= 12;
69         tempFt++;
70     }
71     feet += tempFt;
72     inches += tempIn;
73 }
74
75 void FeetInches::operator -(FeetInches& dis2)
76 {
77     double tempFt = dis2.feet;
78     double tempIn = dis2.inches;
79     feet -= tempFt;
80     inches -= tempIn;
81     while (inches >= 12)
82     {
83         feet -= 12;
84         feet++;
85     }
86     while (inches < 0)
87     {
88         inches += 12;
89         feet--;
90     }
91     feet = absolute(feet);
92 }
93
94 bool FeetInches::operator >(FeetInches& dis2)
95 {
96     if (feet >= dis2.feet && inches > dis2.inches)
97     {
98         return true;
```

```
99     }
100     return false;
101 }
102
103
104 bool FeetInches::operator <(FeetInches& dis2)
105 {
106     if (feet <= dis2.feet && inches < dis2.inches)
107     {
108         return true;
109     }
110     return false;
111 }
112
113 bool FeetInches::operator ==(FeetInches& dis2)
114 {
115     if (feet == dis2.feet && inches == dis2.inches)
116     {
117         return true;
118     }
119     return false;
120 }
121
122 std::istream& operator >>(std::istream& stream, FeetInches& dis)
123 {
124     double tempFt{};
125     double tempIn{};
126     stream >> tempFt;
127     stream >> tempIn;
128
129     dis.feet = tempFt;
130     dis.inches = tempIn;
131     return stream;
132 }
133
134 std::ostream& operator <<(std::ostream& stream, FeetInches& dis) {
135     stream << dis.feet << " feet, " << dis.inches << " inches";
136     return stream;
137 }
138
139 // Member Functions
140
141 double FeetInches::absolute(double d)
142 {
143     if (d < 0)
144         return d * -1;
145     else
146         return d;
147 }
```

```
148
149 void FeetInches::conversion() {
150     while (inches >= 12)
151     {
152         feet -= 12;
153         feet++;
154     }
155     while (inches < 0)
156     {
157         inches += 12;
158         feet--;
159     }
160     feet = absolute(feet);
161 }
```